Chapter 8: Linear Momentum and Collisions

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Lecture 8

Part I

Linear Momentum

Outline of Part I: Linear Momentum

Linear Momentum and Force

Impulse

Why Define Momentum?

Newton's Second Law is commonly written $\vec{F}_{\text{net}} = m\vec{a}$.

But this form **only holds when mass is constant.**

What About Variable-Mass Systems?

- A rocket burns fuel and ejects it — its mass decreases over time.
- Rain falls into an open railcar moving along a track — its mass increases.

We need a more general form of Newton's Second Law.

Linear momentum provides exactly that.

Linear Momentum

Linear Momentum ($\vec{p}$) — SI Unit: $\mathrm{kg} \cdot \mathrm{m/s}$ ($\mathrm{kg} \cdot \mathrm{m/s}$)

The **linear momentum** of an object is the product of its mass and velocity:

$$\vec{p} = m\vec{v}$$

where $\vec{p}$ is momentum, m is mass, and $\vec{v}$ is velocity.

- Momentum is a **vector**: it has both magnitude and direction.
- Its direction is the **same** as the direction of $\vec{v}$.
- Doubling mass or speed doubles momentum.

Note: in 1-D, momentum can be positive or negative, depending on the chosen direction of positive.

Momentum and Newton's Second Law

The Most General Form of Newton's Second Law

Newton's Second Law, in its most general form, is:

$$\vec{F}_{\text{net}}(t) = \frac{\Delta\vec{p}}{\Delta t}$$

The net force on an object equals the *rate of change of its momentum.*

- If mass is constant: $\dfrac{\Delta\vec{p}}{\Delta t} = \dfrac{\Delta(m\vec{v})}{\Delta t} = m\dfrac{\Delta\vec{v}}{\Delta t} = m\vec{a}$
- $\therefore\ \vec{F}_{\text{net}} = m\vec{a}$ is a *special case* valid only when m is constant.

The form $\vec{F}_{\text{net}} = \dfrac{\Delta\vec{p}}{\Delta t}$ is **always valid**.

Example: Comparing Momenta (cp2e_ch8_ex1)

(a) Calculate the momentum of a 110-kg football player running at $8.00\,\frac{\mathrm{m}}{\mathrm{s}}$.

(b) Compare this with the momentum of a hard-thrown $0.410\,\mathrm{kg}$ football at $25.0\,\frac{\mathrm{m}}{\mathrm{s}}$.

(a)
$$\begin{aligned}\vec{p}_{\text{player}} &= m\vec{v} \\ &= 110\,\mathrm{kg} \times 8.00\,\frac{\mathrm{m}}{\mathrm{s}}\end{aligned}$$
$$\boxed{\vec{p}_{\text{player}} = 880\,\mathrm{kg}\cdot\mathrm{m/s}}$$

(b)
$$\begin{aligned}\vec{p}_{\text{ball}} &= m\vec{v} \\ &= 0.410\,\mathrm{kg} \times 25.0\,\frac{\mathrm{m}}{\mathrm{s}}\end{aligned}$$
$$\boxed{\vec{p}_{\text{ball}} = 10.25\,\mathrm{kg}\cdot\mathrm{m/s}}$$

The player's momentum ($880\,\mathrm{kg}\cdot\mathrm{m/s}$) is about **86 times** greater than the football's, despite the ball moving faster.

Check Your Understanding

An object has a certain kinetic energy. If its mass is doubled while its speed is halved, how does its momentum change?

Come to lecture to see the answer.

Check Your Understanding

An object has zero momentum. Does it necessarily have zero kinetic energy?

An object has zero kinetic energy. Does it necessarily have zero momentum?

Come to lecture to see the answer.

Outline of Part I: Linear Momentum

Linear Momentum and Force

Impulse

Impulse

Impulse $\left(\vec{J}\right)$ — SI Unit: $\mathrm{kg \cdot m/s}$ or $\mathrm{N \cdot s}$

Starting from $\vec{F}_{\text{net}} = \dfrac{\Delta \vec{p}}{\Delta t}$, we can rearrange:

$$\vec{J} \equiv \Delta \vec{p} = \vec{F}_{\text{avg}}\, \Delta t$$

Impulse is the change in momentum of an object.
It equals the average net force multiplied by the time interval of contact.

Impulse–Momentum Theorem

$$\vec{J} = \Delta \vec{p} = \vec{p}_f - \vec{p}_0 = \vec{F}_{\text{avg}}\, \Delta t$$

- Large F, small Δt *or*
- Small F, large Δt

can produce the same impulse.

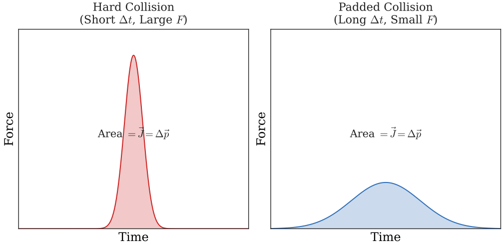
Impulse in Sports and Safety

Increasing Force: Short Contact Time

- Batting, kicking, striking: short Δt $\Rightarrow$ very large F.
- Goal: maximize F to give the ball large Δp.

Decreasing Force: Long Contact Time

- Airbags, padding, foam mats: extend $\Delta t \Rightarrow$ reduce F.
- Same Δp, but spread over a longer time = less injury.



The **area** under a force vs. time graph equals the impulse $\vec{J} = \Delta\vec{p}$.

Padding increases Δt, which reduces $\vec{F}_{\text{avg}}$ for the same $\vec{J}$.

Example: Impulse on a Tennis Ball (cp2e_ch8_ex4)

A tennis ball ($m = 0.057\,\text{kg}$) is struck by a racket. The contact time is $\Delta t = 5.00 \times 10^{-3}\,\text{s}$. The ball's initial horizontal velocity is zero, and its final velocity is $58.0\,\frac{\text{m}}{\text{s}}$. Find the average net horizontal force exerted on the ball.

$$\vec{J} = \Delta\vec{p} = m\vec{v}_f - m\vec{v}_0$$

$$\vec{J} = (0.057\,\text{kg})(58.0\,\tfrac{\text{m}}{\text{s}}) - 0$$

$$\vec{J} = 3.306\,\text{kg}\cdot\text{m/s}$$

$$\vec{F}_{\text{avg}}\Delta t = \vec{J}$$

$$\boxed{\vec{F}_{\text{avg}} = \frac{\vec{J}}{\Delta t} = \frac{3.306\,\text{kg}\cdot\text{m/s}}{5.00\times 10^{-3}\,\text{s}} = 661\,\text{N}}$$

A force of $661\,\text{N}$ acts for only $5\,\text{ms}$.

This is $\sim$10 times the weight of the ball.

Short contact time $\Rightarrow$ enormous average force.

Check Your Understanding

During a collision, can the average force on an object be zero even if the impulse is not zero?

Can the impulse be zero even if the average force is not zero?

Come to lecture to see the answer.

Check Your Understanding

A medicine ball is dropped from height h and bounces off the floor. In a separate trial, the same ball is caught by a person with padded gloves. In both cases, the ball starts with the same speed just before impact.

In which case is the force on the ball larger? Why?

Come to lecture to see the answer.

Part II

Conservation of Momentum

Outline of Part II: Conservation of Momentum

Internal and External Forces

Consider a **system** of objects (e.g., two colliding carts).

Internal Forces

- Forces *between* objects inside the system.
- Example: the contact force between two colliding carts.
- By Newton's 3rd Law, internal forces always come in equal and opposite pairs $\Rightarrow$ they cancel out in the total.

External Forces

- Forces from *outside* the system acting on objects inside it.
- Example: friction from the floor, gravity, a push from a hand.
- External forces *can* change the total momentum of the system.

Conservation of Linear Momentum

Law of Conservation of Linear Momentum

For an **isolated system** (no net external force), the total momentum is constant:

$$\Delta \vec{p}_{\text{sys}} = 0 \qquad \therefore \qquad \vec{p}_{\text{sys, after}} = \vec{p}_{\text{sys, before}}$$

For Two Objects

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = m_1 \vec{v}_1' + m_2 \vec{v}_2'$$

where primes ($'$) denote *after* the interaction.

- This holds **regardless** of the type of collision (elastic or inelastic).
- Conservation of momentum is a *consequence* of Newton's 3rd Law applied to all internal forces.

Example: Recoil Velocity (cp2e_ch8_ex5)

A $1.80\,\mathrm{kg}$ rifle fires a $30.0\,\mathrm{g}$ bullet at $500\,\frac{\mathrm{m}}{\mathrm{s}}$. The rifle and shooter together have a mass of $90.0\,\mathrm{kg}$. The system is initially at rest. Find the recoil velocity of the rifle-shooter system.

$$
\begin{aligned}
p_{\text{before}} &= p_{\text{after}} \\
0 &= m_{\text{bullet}}\, v'_{\text{bullet}} + m_{\text{rifle}}\, v'_{\text{rifle}} \\
v'_{\text{rifle}} &= -\frac{m_{\text{bullet}}\, v'_{\text{bullet}}}{m_{\text{rifle}}} \\
v'_{\text{rifle}} &= -\frac{(0.030\,\mathrm{kg})(500\,\frac{\mathrm{m}}{\mathrm{s}})}{90.0\,\mathrm{kg} + 1.80\,\mathrm{kg}}
\end{aligned}
$$

$$\boxed{v'_{\text{rifle}} = -0.163\,\frac{\mathrm{m}}{\mathrm{s}}}$$

The negative sign means the rifle recoils *opposite* to the bullet's direction.

The system's total momentum remains zero, as expected.

Example: Finding Mass from Momentum Conservation

Object 1 (unknown mass m_1) is initially at rest. Object 2 ($m_2 = 5.00\,\text{kg}$) moves at $\vec{v}_2 = 3.00\,\frac{\text{m}}{\text{s}}$ in the $+x$ direction. After a collision: $\vec{v}_2' = -2.00\,\frac{\text{m}}{\text{s}}$ and $\vec{v}_1' = 1.00\,\frac{\text{m}}{\text{s}}$ (both in x). Find m_1.

$$m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$$

$$0 + (5.00)(3.00) = m_1(1.00) + (5.00)(-2.00)$$

$$15.0 = m_1 - 10.0$$

$$\boxed{m_1 = 25.0\,\text{kg}}$$

Notice that the heavier object ($m_1 = 25\,\text{kg}$) barely moves ($1.00\,\frac{\text{m}}{\text{s}}$), while the lighter one ($5\,\text{kg}$) bounces back.

Check Your Understanding

Two ice skaters push off from each other. Skater A has mass 60.0 kg and moves at $1.50\,\frac{\mathrm{m}}{\mathrm{s}}$ to the right after the push. Skater B has mass 80.0 kg.

What is Skater B's velocity after the push? (Assume the system was initially at rest.)

Come to lecture to see the answer.

Check Your Understanding

In a game of pool (billiards), a player strikes the cue ball, which hits an initially stationary ball.

Can both balls be at rest immediately after the collision?

Can one ball be at rest? Explain.

Come to lecture to see the answer.

Part III

Collisions

Outline of Part III: Collisions

Types of Collisions

All collisions conserve momentum (for an isolated system).
Collisions are classified by what happens to **kinetic energy**.

Type	**Momentum**	**Kinetic Energy**
Elastic	Conserved	Conserved
Inelastic	Conserved	Not conserved (KE lost)
Perfectly Inelastic	Conserved	Maximum KE lost

- "Lost" KE is converted to heat, sound, or deformation — it is not truly destroyed, just transformed.
- **Perfectly inelastic:** objects *stick together* after the collision

Check Your Understanding

Is it possible for two objects to collide and *gain* kinetic energy during the collision?

Give an example or explain why not.

Come to lecture to see the answer.

Outline of Part III: Collisions

Elastic Collisions in One Dimension

Definition

An **elastic collision** conserves both momentum *and* kinetic energy:

$$m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$$

$$\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = \frac{1}{2} m_1 v_1'^2 + \frac{1}{2} m_2 v_2'^2$$

Resulting Velocity Formulas

Solving the two equations simultaneously gives:

$$v_1' = \frac{m_1 - m_2}{m_1 + m_2} v_1 + \frac{2m_2}{m_1 + m_2} v_2$$

$$v_2' = \frac{2m_1}{m_1 + m_2} v_1 - \frac{m_1 - m_2}{m_1 + m_2} v_2$$

Special Cases: Elastic Collisions

Equal Masses ($m_1 = m_2$)

$v_1' = v_2, \quad v_2' = v_1$

Objects *exchange* velocities.

If ball 2 is at rest: ball 1 **stops**, ball 2 **moves at** v_1.

Much Heavier Target ($m_2 \gg m_1$)

$v_1' \approx -v_1, \quad v_2' \approx 0$

Light object bounces back at nearly the same speed; heavy target barely moves (like bouncing off a wall).

Much Heavier Striker ($m_1 \gg m_2$)

$v_1' \approx v_1, \quad v_2' \approx 2v_1$

Heavy object barely slows; light one flies off at $\sim 2v_1$.

All three cases follow from the formulas on the previous slide.

Example: Elastic Collision (cp2e_ch8_ex6)

Two carts collide elastically on a frictionless track. Cart 1 ($m_1 = 0.250\,\text{kg}$) moves at $0.800\,\frac{\text{m}}{\text{s}}$ and cart 2 ($m_2 = 0.500\,\text{kg}$) is initially at rest ($v_2 = 0$).
Find v_1' and v_2'.

$$v_1' = \frac{m_1 - m_2}{m_1 + m_2}\, v_1 = \frac{0.250 - 0.500}{0.750}(0.800)$$

$$\boxed{v_1' = -0.267\,\frac{\text{m}}{\text{s}}}$$

$$v_2' = \frac{2m_1}{m_1 + m_2}\, v_1 = \frac{2(0.250)}{0.750}(0.800)$$

$$\boxed{v_2' = 0.533\,\frac{\text{m}}{\text{s}}}$$

Cart 1 bounces back (v_1' is negative).
Cart 2 moves forward at $0.533\,\frac{\text{m}}{\text{s}}$.

Check KE:
Before: $\frac{1}{2}(0.250)(0.800)^2 = 0.0800\,\text{J}$
After: $\frac{1}{2}(0.250)(0.267)^2 + \frac{1}{2}(0.500)(0.533)^2 \approx 0.0800\,\text{J}$ ✓

Example: Verifying an Elastic Collision

Object 1 ($m_1 = 25.0\,\text{kg}$, at rest) is struck by Object 2 ($m_2 = 5.00\,\text{kg}$, $v_2 = 3.00\,\frac{\text{m}}{\text{s}}$). After the collision: $v_1' = 1.00\,\frac{\text{m}}{\text{s}}$, $v_2' = -2.00\,\frac{\text{m}}{\text{s}}$.

Verify that this is an elastic collision.

Before

$$\begin{aligned} KE_{\text{before}} &= \tfrac{1}{2}(25.0)(0)^2 + \tfrac{1}{2}(5.00)(3.00)^2 \\ &= 0 + 22.5\,\text{J} \end{aligned}$$

After

$$\begin{aligned} KE_{\text{after}} &= \tfrac{1}{2}(25.0)(1.00)^2 + \tfrac{1}{2}(5.00)(2.00)^2 \\ &= 12.5 + 10.0 = 22.5\,\text{J} \end{aligned}$$

$KE_{\text{before}} = KE_{\text{after}} = 22.5\,\text{J}$ ✓ This **is** an elastic collision.

Check Your Understanding

Two identical balls ($m_1 = m_2 = m$) collide head-on elastically. Ball 1 moves at $+v$ and Ball 2 moves at $-v$ before the collision.

What are the velocities of each ball after the collision?

Come to lecture to see the answer.

Outline of Part III: Collisions

Inelastic Collisions

Inelastic Collision

An **inelastic collision** conserves momentum but *not* kinetic energy.

Some kinetic energy is converted to internal energy (heat, sound, deformation).

$$m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2' \qquad \text{(still holds)}$$

Perfectly Inelastic Collision

Objects **stick together** after the collision: $v_1' = v_2' \equiv v'$.

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2)\, v' \qquad \Rightarrow \qquad v' = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

The maximum kinetic energy is lost in a perfectly inelastic collision.

Example: Perfectly Inelastic Collision (cp2e_ch8_ex7)

A 1000 kg car moving at 9.00 $\frac{\text{m}}{\text{s}}$ collides with a 3000 kg truck at rest. They crumple together. Find their common velocity v' and the fraction of KE lost.

$$v' = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

$$= \frac{(1000)(9.00) + (3000)(0)}{1000 + 3000}$$

$$\boxed{v' = 2.25\,\frac{\text{m}}{\text{s}}}$$

$$KE_{\text{before}} = \tfrac{1}{2}(1000)(9.00)^2 = 40\,500\text{ J}$$

$$KE_{\text{after}} = \tfrac{1}{2}(4000)(2.25)^2 = 10\,125\text{ J}$$

$$\%\text{ lost} = \frac{40500 - 10125}{40500} \approx \boxed{75\%}$$

Check Your Understanding

Two clay balls collide and stick together. Ball 1 ($m_1 = 3.0\,\text{kg}$) moves at $4.0\,\frac{\text{m}}{\text{s}}$ to the right. Ball 2 ($m_2 = 3.0\,\text{kg}$) moves at $4.0\,\frac{\text{m}}{\text{s}}$ to the left.

What is the final velocity of the combined lump? How much KE was lost?

Come to lecture to see the answer.

Check Your Understanding

After a perfectly inelastic collision, is it possible that both objects end up at rest? Under what condition?

Is it possible that one object ends up at rest?

Come to lecture to see the answer.

Outline of Part III: Collisions

Types of Collisions

Elastic Collisions in One Dimension

Inelastic Collisions in One Dimension

Collisions of Point Masses in Two Dimensions

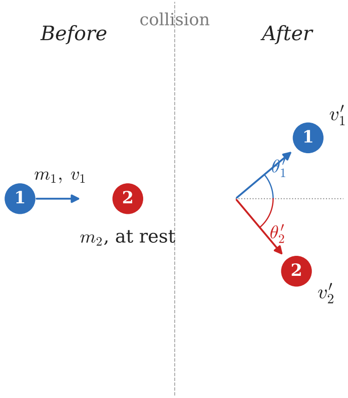
Collisions in Two Dimensions: Setup

In 2-D, momentum is a vector $\Rightarrow$ conserve x- **and** y**-components separately**.

Conservation Equations for Two Objects

$$
\begin{aligned}
x: &\quad m_1 v_{1x} + m_2 v_{2x} = m_1 v'_{1x} + m_2 v'_{2x} \\
y: &\quad m_1 v_{1y} + m_2 v_{2y} = m_1 v'_{1y} + m_2 v'_{2y}
\end{aligned}
$$

- This gives **two** equations (x and y).
- Elastic collisions: KE equation adds a third.
- Typically need one post-collision angle or speed to solve.



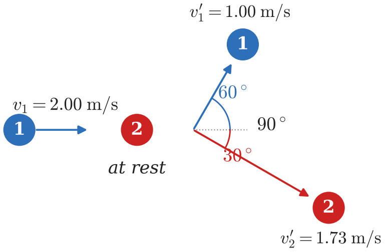
2-D Elastic Collision: Equal Masses

Key Result for Equal-Mass Elastic Collisions

When $m_1 = m_2$ and ball 2 is initially at rest, the two balls scatter at **right angles** to each other after the collision:

$$\theta_1' + \theta_2' = 90^\circ$$

This right-angle result is why billiard balls often scatter at nearly 90° — billiard balls have nearly equal masses.



Equal-mass elastic: $\theta_1' + \theta_2' = 90^\circ$

Example: 2-D Elastic Collision (cp2e_ch8_ex7)

Two identical billiard balls: ball 1 moves at $v_1 = 2.00\,\frac{\mathrm{m}}{\mathrm{s}}$ in the $+x$ direction; ball 2 is at rest. After an elastic collision, ball 2 moves at $\theta_2' = 30.0^\circ$ below the x-axis. Find v_1', v_2', and θ_1'.

Using $\theta_1' + \theta_2' = 90^\circ$ (equal masses)

$$\theta_1' = 90^\circ - 30^\circ = \boxed{60.0^\circ}$$

$$x: \quad 2.00 = v_1' \cos 60^\circ + v_2' \cos 30^\circ = \frac{v_1'}{2} + \frac{\sqrt{3}\,v_2'}{2}$$

$$y: \quad 0 = v_1' \sin 60^\circ - v_2' \sin 30^\circ = \frac{\sqrt{3}\,v_1'}{2} - \frac{v_2'}{2}$$

From y: $v_2' = \sqrt{3}\,v_1'$

Substitute into x:

$$2.00 = \frac{v_1'}{2} + \frac{\sqrt{3}(\sqrt{3}\,v_1')}{2} = 2v_1'$$

$$\boxed{v_1' = 1.00\,\frac{\mathrm{m}}{\mathrm{s}}}$$

$$\boxed{v_2' = \sqrt{3} \approx 1.73\,\frac{\mathrm{m}}{\mathrm{s}}}$$

Check Your Understanding

Two identical billiard balls undergo an elastic collision. Ball 1 moves in the $+x$ direction; ball 2 is at rest. After the collision, ball 2 is observed to move at $45°$ below the x-axis.

What angle does ball 1 make with the $+x$ axis after the collision?

Come to lecture to see the answer.

Part IV

Rocket Propulsion

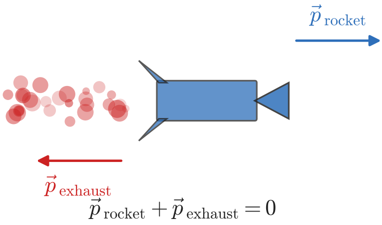
Outline of Part IV: Rocket Propulsion

Rocket Propulsion: Momentum Conservation

How Does a Rocket Accelerate in Empty Space?

There is nothing to push against — no ground, no air.

A rocket propels itself by **ejecting mass** (exhaust gas) at high speed.



By conservation of momentum:

- Before: rocket + fuel at rest $\Rightarrow p = 0$.
- After: exhaust goes backward, rocket goes forward.
- $\Delta p_{\text{rocket}} = -\Delta p_{\text{exhaust}}$

Newton's 3rd Law: rocket pushes gas backward; gas pushes the rocket forward.

Rocket Propulsion: The Equation

Rocket Acceleration

The acceleration of a rocket is:

$$a_r = \frac{v_e}{m}\frac{\Delta m}{\Delta t} - g$$

where:

- v_e = **exhaust speed** (speed of gas relative to the rocket), always positive
- m = **current mass** of the rocket (decreasing as fuel burns)
- $\dfrac{\Delta m}{\Delta t}$ = **rate of mass ejection** (positive; the rate at which mass is lost)
- g = gravitational acceleration ($9.8\ \frac{\mathrm{m}}{\mathrm{s}^2}$ near Earth's surface)

Understanding the Rocket Equation

Thrust

The term $\frac{v_e}{m}\frac{\Delta m}{\Delta t}$ is the **thrust acceleration**:

- Thrust $= v_e \frac{\Delta m}{\Delta t}$ (in Newtons)
- As fuel burns, m decreases, so thrust acceleration *increases* over time for constant thrust.

Gravity Term

The $-g$ term represents the downward gravitational acceleration the rocket must overcome.

In outer space far from a planet: $g \approx 0$, so $a_r = \frac{v_e}{m}\frac{\Delta m}{\Delta t}$.

The rocket accelerates as long as **thrust** > **weight**: $v_e \frac{\Delta m}{\Delta t} > mg$.

Example: Rocket Acceleration (cp2e_ch8_ex8)

A large rocket has an initial mass of $m = 2.80 \times 10^6$ kg, exhaust speed $v_e = 2.40 \times 10^3\,\frac{\text{m}}{\text{s}}$, and ejects mass at $\dfrac{\Delta m}{\Delta t} = 1.40 \times 10^4\,\frac{\text{kg}}{\text{s}}$.
Find the initial acceleration of the rocket.

$$a_r = \frac{v_e}{m}\frac{\Delta m}{\Delta t} - g$$

$$= \frac{(2.40 \times 10^3\,\frac{\text{m}}{\text{s}})(1.40 \times 10^4\,\frac{\text{kg}}{\text{s}})}{2.80 \times 10^6\,\text{kg}} - 9.80\,\frac{\text{m}}{\text{s}^2}$$

$$= 12.0\,\frac{\text{m}}{\text{s}^2} - 9.80\,\frac{\text{m}}{\text{s}^2}$$

$$\boxed{a_r = 2.20\,\frac{\text{m}}{\text{s}^2}}$$

The thrust produces $12.0\,\frac{\text{m}}{\text{s}^2}$, but gravity subtracts $9.80\,\frac{\text{m}}{\text{s}^2}$, leaving only $2.20\,\frac{\text{m}}{\text{s}^2}$ net upward.

As fuel burns, m decreases $\Rightarrow a_r$ increases.

Check Your Understanding

A rocket is in deep space, far from any planet (so $g \approx 0$). If the rocket doubles its exhaust speed while keeping the same mass-ejection rate and same current mass, how does its acceleration change?

Come to lecture to see the answer.

Check Your Understanding

A rocket burns fuel at a constant rate. As the mission continues, fuel is used up and the rocket's mass decreases.

Does the rocket's acceleration increase, decrease, or stay the same over time? Assume constant exhaust speed and constant $\Delta m/\Delta t$, and that gravity is negligible.

Come to lecture to see the answer.

Chapter Summary

Key Equations

- $\vec{p} = m\vec{v}$
- $\vec{F}_{\text{net}} = \dfrac{\Delta\vec{p}}{\Delta t}$
- $\vec{J} = \Delta\vec{p} = \vec{F}_{\text{avg}}\,\Delta t$
- $\Delta\vec{p}_{\text{sys}} = 0$ (isolated system)
- $v' = \dfrac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$ (PIC)
- $a_r = \dfrac{v_e}{m}\dfrac{\Delta m}{\Delta t} - g$

Key Concepts

- Momentum is always conserved in an isolated system.
- KE is only conserved in *elastic* collisions.
- Impulse = area under F vs. t graph.
- In 2-D, conserve x and y separately.
- Rockets work by momentum conservation — no medium required.